

## Topic 2.5 - Exponential Function Context and Data Modeling

### Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

#### Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

#### Learning focus

- Construct exponential models from percent rates, two points, transformations, or regression output.
- Use exponential models to predict outputs and solve for inputs.
- Interpret parameters, equivalent time scales, domains, units, and model limitations.

### Section A: Foundation and representation

#### Question 1

**Modeling****Calculator permitted**

A fund begins with 8500 QAR and grows by 4.2% annually. Write a model and find the balance after 8 years.

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**Question 2**

Modeling

Calculator permitted

A medication amount is 240 mg initially and decreases by 18% each hour. Find the amount after 6 hours and the first whole hour when it is below 50 mg.

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**Question 3**

Core skill

No calculator

A population is 1800 at year 2 and 4050 at year 4. Construct an exponential model in the form  $P(t) = ab^t$ .

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**Question 4**

Equivalent intervals

Calculator permitted

A quantity triples every 11 days. Write a model with initial value  $Q_0$ , then find the factor for one day.

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**Question 5**

Characteristic time

Calculator permitted

A radioactive sample has mass 96 g and half-life 14 years. Find the mass after 35 years.

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**Question 6**

Interpretation

No calculator

A machine's efficiency is modeled by  $E(t) = 72 + 18(0.65)^t$ . Interpret all three numerical parameters and state the long-run efficiency.

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**Section B: Application, comparison, and error analysis****Question 7**

Shifted model

No calculator

A temperature approaches  $24^\circ\text{C}$  and satisfies  $T(0) = 88$  and  $T(4) = 50.2144$ . Construct the model.

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**Question 8**

Solve for input

Calculator permitted

Use  $P(t) = 1500(1.12)^t$  to solve for the time when  $P = 3000$ . Give an exact logarithmic expression and a decimal approximation.

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**Question 9**

Error analysis

No calculator

A student fits an exponential regression to six data points and reports  $R^2 = 0.998$ . Explain why this does not prove that the underlying process is exponential.

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**Question 10**

Equivalent forms

Calculator permitted

Two models for the same data are  $A(t) = 400(1.09)^t$  and  $B(t) = 400(1.6771)^{t/6}$ . Determine whether they are approximately equivalent.

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**Question 11**

Model limitation

No calculator

A town population model predicts  $P(20) = 180000$ , but the physical capacity is estimated at 120000. What does this discrepancy say about the model?

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**Question 12**

Comparison

Calculator permitted

Construct two different exponential models that both have value 50 at  $t = 0$  but one grows by 12% per year and the other doubles every 6 years. Compare their values at  $t = 6$ .

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**Section C: AP-style multiple choice****MCQ 1**

No calculator

A quantity decreases by 7% per month. Which factor belongs in an exponential model?

- A. 1.07
- B. 0.93
- C. 0.07
- D.  $-0.07$

**MCQ 2**

No calculator

A model doubles every 4 hours. Which formula is correct?

- A.  $P(t) = P_0 2^{4t}$
- B.  $P(t) = P_0 4^{t/2}$
- C.  $P(t) = P_0 2^{t/4}$
- D.  $P(t) = P_0 (1/2)^{t/4}$

**MCQ 3**

No calculator

For  $T(t) = 18 + 70(0.8)^t$ , what does 70 represent?

- A. The limiting temperature
- B. The initial temperature
- C. The initial difference from the limit
- D. The percent decrease

**MCQ 4**

No calculator

Which statement is the strongest reason not to extrapolate an exponential population model indefinitely?

- A. Exponential functions are never used for populations.
- B. The graph has a horizontal asymptote.
- C. Resource limits can invalidate the constant proportional-growth assumption.
- D. All populations eventually become linear.

**Section D: Free-response reasoning****FRQ 1 - Exponential growth model**

Calculator permitted

6 points

A research colony contains 750 organisms at day 0 and 1125 organisms at day 3. (a) Construct an exponential model. (b) Interpret the three-day and one-day growth factors. (c) Predict the population at day 10. (d) Determine when the model first predicts more than 4000 organisms and discuss one limitation.

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**FRQ 2 - Shifted decay model**

Calculator permitted

6 points

A liquid cools toward  $22^{\circ}\text{C}$ . Its temperature is  $94^{\circ}\text{C}$  at time 0 and  $46.576^{\circ}\text{C}$  at time 3. (a) Construct a model  $T(t) = 22 + Ab^t$ . (b) Interpret  $A$  and  $b$ . (c) Find the temperature at time 7. (d) Solve for the time when the temperature is  $30^{\circ}\text{C}$  and comment on whether the model ever reaches  $22^{\circ}\text{C}$ .

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